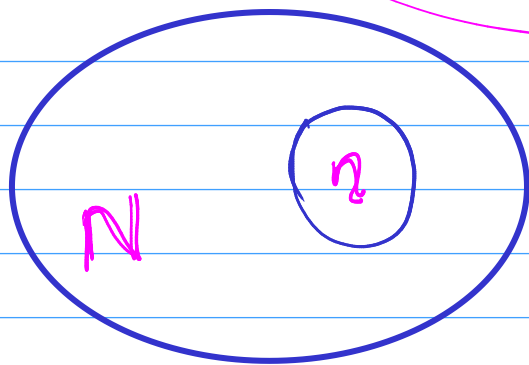


Sampling Techniques



Selection of samples from the population data.

A = Simple Random Sampling ✓

↳ If we need to find churn Rate of the employees, we collect the feedback form randomly.

B. Stratified Sampling

⇒ We collect samples using some strategy.

ex ⇒ Loan Calls (Income Range)

ex ⇒ Election Commission feedback
(Age ⇒ 18+)

ex ⇒ Gender .

C. Systematic Sampling

→ Samples collected by system

ex :- nth no. System, (6, 16, 26, 36)
= Regular wise system.
(1, 2, 3, 4, 5)
(ving)

D. Convenience Sampling

ex → Credit Card

ex → Blood Camp

E. Purposive Sampling

→ Gov, Research sector

→ Covid19 / Cylinder

F. Cluster Sampling

Div

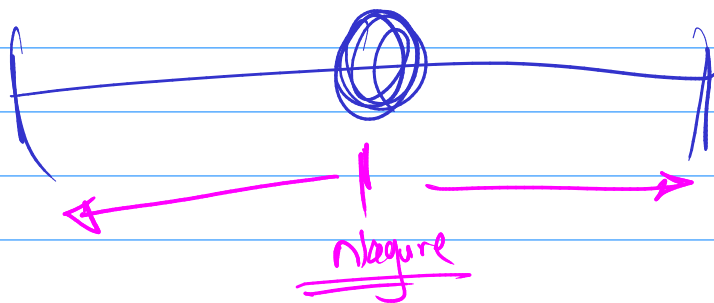
Khandha district

Nag
yar. Am

Tah → seloo ✓
→ Khandha ✓
→ Aruo

⇒ Avg. height of Indian people?

① Measure of Central Tendency



mean
median
mode
Trimmed mean
weighted mean

② Measure of Dispersion — Spread

① Range ⇒ The difference between the maximum & minimum value.

ex $\Rightarrow 17, 18, 19, 20, 21, 17, 19, 20, 21, 20, 20$

mean $\Rightarrow 18, 19$

Range = $21 - 17$ = 4

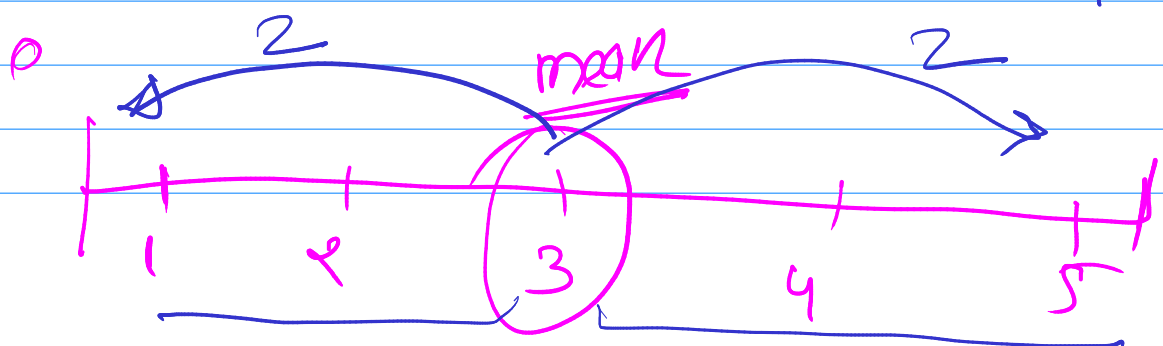
km = 80k

$\hookrightarrow$ It is Robust to Outlier
affected

② Variance $\sigma^2 \Rightarrow$ Population (σ^2) Sample (s^2)
Sigma

$\sigma^2 \Rightarrow$ It is the average of the Squared
Difference between each data point and the
mean

<u>data point</u>		<u>$x_i - \mu$</u>	<u>$(x_i - \mu)^2$</u>	$\frac{\text{mean}}{(\mu)} = \frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$
x_1	1	$1-3 = -2$	4	
x_2	2	-1	1	$\sigma^2 = \frac{4+1+0+1+4}{5}$ $= \frac{10}{5} = 2$
x_3	3	0	0	
x_4	4	1	1	
x_5	5	2	4	



③ Standard Deviation (SD)

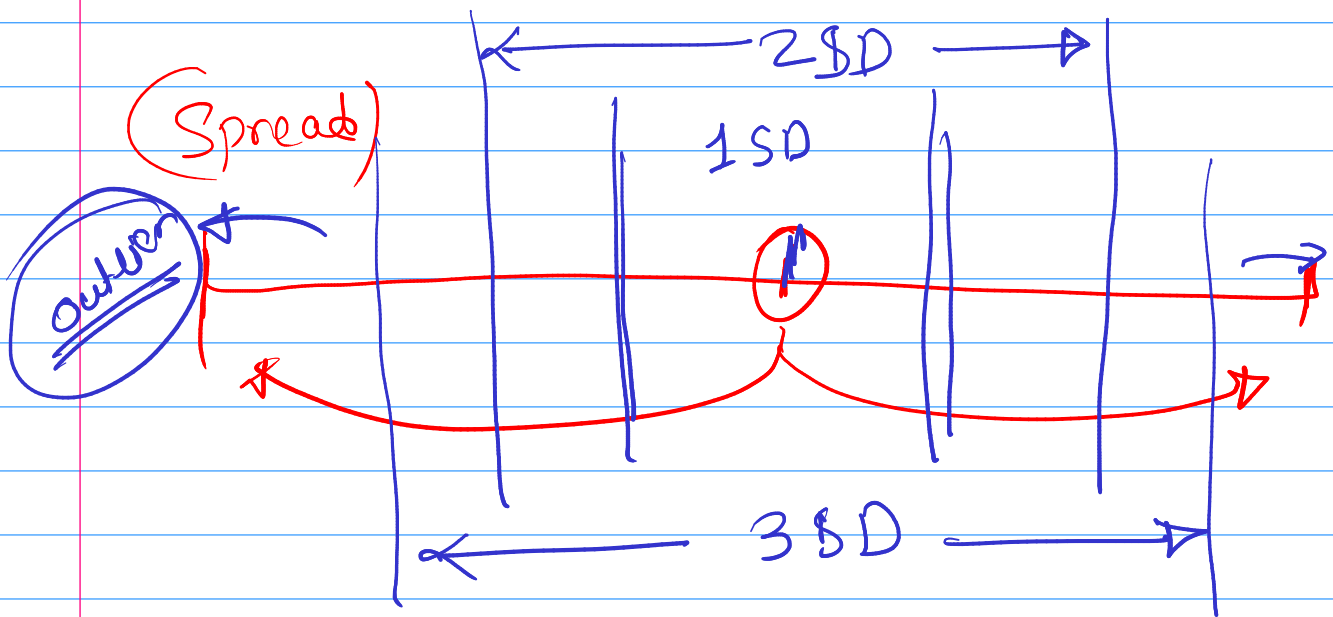
→ It is the square root of Variance

$$SD = \sqrt{\sigma^2}$$

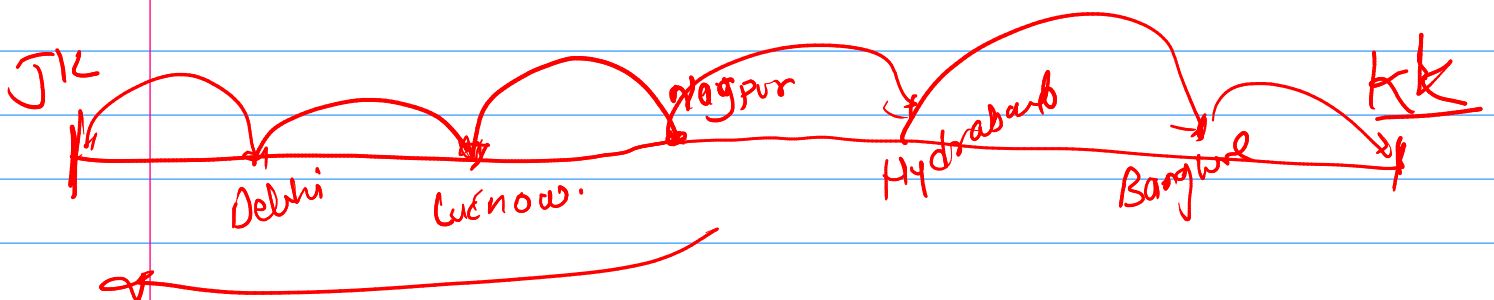
$$\cancel{km^2} = \sqrt{\cancel{km^2}}$$

unit square Cancel

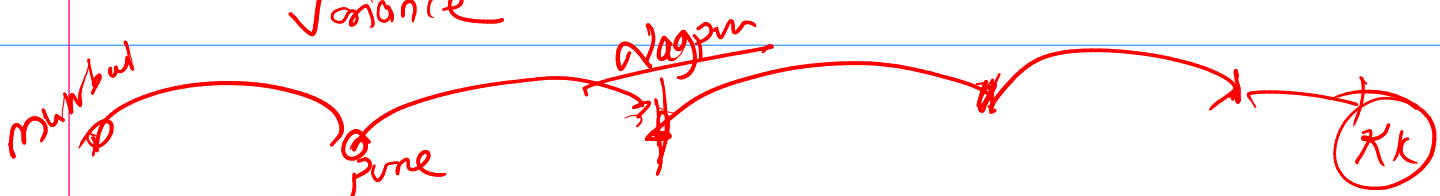
moab



Ex → Airport



Variance



④ Coefficient of Variation = CV

↓ medical

g → It is the ratio of Standard deviation to the mean expressed as percentage.

$$CV = \left(\frac{SD}{\mu} \right) \times 100\%$$

why?

Salary class	Year of Experience (year)
1	1
2	2
3	3
4	4
5	5

g mg

5% 0.3% → CV